

CROSSING THE CHASM

Marketing and Selling Disruptive Products
to Mainstream Customers

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The Big “So What”

Launching a high-tech product can be extremely challenging, and failure comes with the loss of significant time and resources. **This book provides proven frameworks and strategies to successfully bring new, cutting-edge products to market, with decision-making guidelines for investors, engineers, marketers and managers in the high-tech community.**



Introduction

A disruptive or discontinuous innovation is one that requires people to change their existing behaviors, whereas a sustaining or continuous innovation involves a upgrade of existing products. A bigger or higher-resolution TV screen is a continuous innovation while a 3D-TV (which requires special viewing glasses) is a discontinuous innovation.

Every industry will face disruptive changes at some point, though high-tech industries experience them on a regular basis. Hence, it's crucial to have a systematic approach to market new high-tech products.

Companies are especially vulnerable when they're trying to cross the chasm between the small minority of early customers and the large majority of mainstream customers. **In this summary, we'll introduce the chasm and the challenges it presents, before explaining how you can successfully cross it to achieve growth and profits.**

KEY QUOTES

“The point of greatest peril in the development of a hightech market lies in making the transition from an early market dominated... to a mainstream market.”



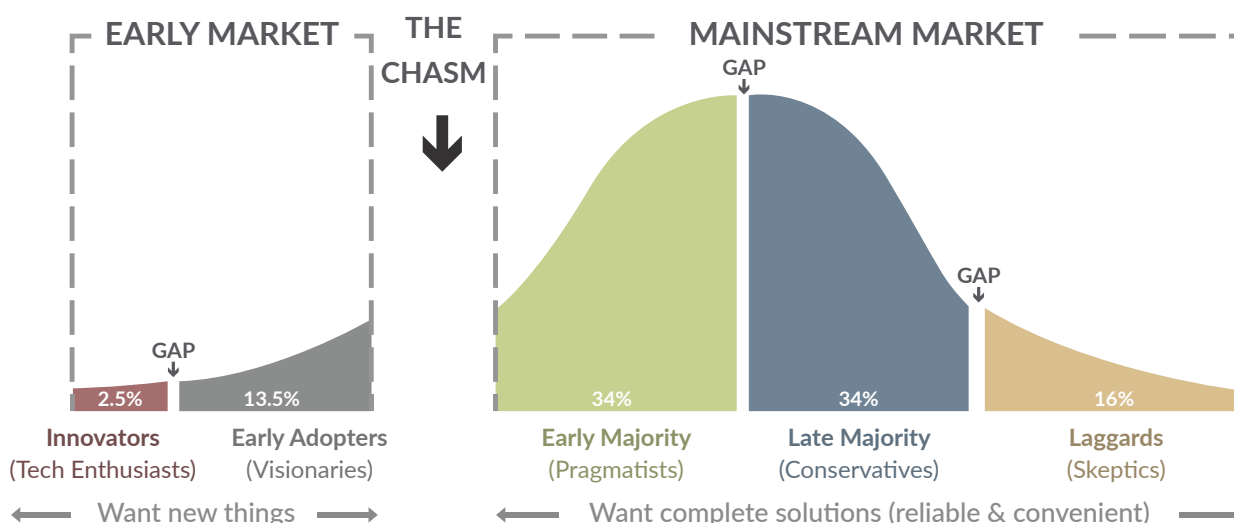
Understanding the Chasm



High-Tech Adoption and Marketing

 The Technology Adoption Life Cycle explains how new products are accepted. Generally, technology adoption follows a bell curve, with 5 distinct consumer groups differentiated by their psychographic profiles and attitudes to new technology:

- Innovators (tech enthusiasts) are passionate about new technology. They actively pursue and explore the latest technology to learn about them.
- Early adopters (visionaries) embrace new technologies as a means to achieve a broader vision. They're good at explaining the potential benefits to others. John F. Kennedy championed the launch of the U.S. space program, while Henry Ford championed the mass-production of automobiles.
- The early majority (pragmatists) are more cautious and realistic. They're only willing to adopt technology that's proven and convenient.
- The late majority (conservatives) are uncomfortable with new technology. They'll adopt it only when most people are already using it, just so they won't get left behind.
- Laggards (skeptics) don't want to be associated with new technology *at all* and may only use it only unknowingly.



KEY QUOTES

"Crossing this chasm...is how high-tech fortunes are made; failure in the attempt is how they are lost."

»»» The High-Tech Marketing Model is about developing the market by sequentially addressing the 5 groups above from left to right. It has been widely applied by many companies from Microsoft to Cisco, Dell, Google, HP and Oracle.

- This model works because each group paves the way for the next group, e.g. innovators' endorsement helps to win over the early adopters, who then pave the way for the early majority and late majority. The Apple iPad was demo-ed to techies at MacWorld before its launch in 2009. Visionary executives started to use them, gradually influencing other members of their organizations. As the number of apps and users increased, even kids and grandparents started to use the iPad. Within 5 years, iPads had become a part of people's everyday lives.
- However, each of the 5 groups have different buying objective and expectations, so what appeals to 1 group can alienate another. Such gaps between the groups can break your marketing momentum. Specifically:
 - (i) There are cracks (a) between the enthusiasts and visionaries, and (b) between the pragmatists and conservatives; and
 - (ii) There's a chasm between visionaries and pragmatists due to their conflicting goals. Visionaries want major changes to leapfrog their competitors, whereas pragmatists want to improve existing operations with minimal disruption.



Marketing to Early vs Mainstream Markets



Marketing is not about creating an illusion or hype. It's about taking concrete actions to define, grow, maintain and defend your target markets. You must define the target segment for each set of products/services. Within each segment, customers must (i) share the same needs/wants and (ii) act as references for each other in their purchasing decisions. If 2 customers can't be cross-referenced, then they are not in the same market, e.g. a doctor won't buy an oscilloscope after hearing how an engineer uses it for his tech research.

KEY QUOTES

"Marketing's purpose...is to develop and shape something that is real, and not...to create illusions."

"No meaningful marketing program can be implemented across a set of customers who do not reference each other."



To succeed in early markets, you need a breakthrough technology product that (i) can appeal to the technology enthusiast *and* (ii) can offer a new/compelling application for the visionary.

- Tech enthusiasts are easy to serve if you have a truly new/great technology and don't need to make much money from them.

	 Innovators / Tech Enthusiasts
 Profile	• Have the passion and mandate to explore new tech • Willing to invest time/energy to get new tech to work. • Willing to tolerate bugs, slow speed & missing functions
 Meet their needs	• Let them be the first to try new stuff with no/low costs • Give honest answers to their technical questions • Invite and adopt their inputs
 To reach them	• Direct email or advertisements via targeted channels • Offer free info/demos; provide novel & factual info (not marketing hype)
 Vital because	• A valuable sounding board to help you refine your product • Gatekeepers/access to key decision-makers

- Visionaries are hard to please, but can give you the big break you need in terms of revenues and visibility.

	 Early Adopters / Visionaries
 Profile	• Willing to take risks and make huge investments in new technology to realize an ambitious dream/vision • Seek major breakthroughs (not minor improvements)
 Meet their needs	• Focus on the strategic leap made possible by your tech • Break down ambitious goals into smaller projects • Use concrete milestones to show progress/results
 To reach them	• Nurture the right relationships and they will find you (not vice versa)
 Vital because	• Opportunity for publicity and high-profile references • Each feature developed can be used as a product extension for another company

- Tips and pitfalls to avoid in early markets:
 - (i) Bridge the crack between the tech enthusiast and visionary: articulate how the Returns on Investment (ROI) on your technology can justify the risks.

KEY QUOTES

"Enthusiasts are like kindling: They help start the fire."

"In business, technology enthusiasts are the gatekeepers for any new technology."

"Visionaries are the ones who give high-tech companies their first big break. It is hard to plan for them in marketing programs, but it is even harder to plan without them."

"Understand their dream, and you will understand how to market to them."

"If it is not, in fact, a breakthrough product, then it is never going to create an early market."

(ii) Don't sell to the visionary before your product is ready, since they're impatient for results.

(iii) Don't be overly-ambitious: select a market that matches your size/resources. It's better to focus on a simpler problem you can solve than to have an unhappy customer.

 Real growth and profits come from the mainstream markets since the early and late majority jointly make up 2/3^{rds} of the entire market. Enter the market with the pragmatists, sustain your leadership position, then move on to the conservatives.

- Pragmatists prefer established/proven solutions, so it's hard for new players to win their business. But once you get in, it's much easier to grow your volume and improve your price/support to get even more sales.

	 Early Majority / Pragmatists
 Profile	• Seek predictable, incremental progress • Prefer proven approaches and standards
 Meet their needs	• Prefer market leaders with a proven track record + can offer reliability, quality and strong support. • Want at least some competition to get the best price & options. Willing to pay a small premium for quality.
 To reach them	• Make sure they hear about you from various industry & media sources, get good testimonials from relevant companies and partner with their trusted vendors.
 Vital because	• A source of valuable references • Loyal support & repeat business once you win their trust

- Conservatives are highly risk-averse and are afraid of high-tech stuff. They prefer status quo and will invest only in mature, commoditized products with low prices.

	 Late Majority / Conservatives
 Profile	• Risk-averse and price-sensitive • Prefer mature products that're convenient and risk-free
 Meet their needs	• Offer standard, heavily-discount bundles for 1 specific need • Include only features that're familiar to them and convenient to use. Remove extra or fancy features.

KEY QUOTES

"If the goal of visionaries is to take a quantum leap forward, the goal of pragmatists is to make a percentage improvement—incremental, measurable, predictable progress."

"The key is to focus on convenience rather than performance, user experience rather than feature sets."

 To reach them	• Use low-cost distribution channels e.g. “as-a-service” product via the internet
 Vital because	• Provide low margins but high volumes • Ideal for older tech elements that can be sold at low cost

- Tip: Once you’ve penetrated the mainstream market, stay ahead of your competitors via incremental product innovations or upgrades. Focus on building predictability, service and stability.
- Laggards make up about 1/6th of the curve: they won’t buy high-tech products and may even block them. Instead of ignoring them, address their concerns—that new technologies are risky and unproven—so they can’t block the sales.



Crossing the Chasm



Companies are especially vulnerable during the chasm period:

- At this point, the early market is still digesting the new technology and the mainstream market is still not yet ready for it. Many companies make the fatal mistake of trying to expand rapidly during the chasm instead of conserving their resources. They try to push pricey and buggy products to mainstream customers (who want cheap and bug-free products), and they continue to invest in high-tech R&D (when mainstream customers want proven, stable solutions).
- It’s also hard to enter a mainstream market since the established incumbents will try to block you. Don’t make the mistake of getting distracted by the diverse opportunities and spreading your resources too thinly.
- **The key to successfully crossing the chasm is to start by focusing singularly on 1 strategic market niche.** Quickly establish a solid base (with references, procedures, marketing materials) and use it to spread to adjacent markets. We’ll now zoom in on the 4 steps to enter and conquer a mainstream market.

KEY QUOTES

“We need to pick our spots carefully, attack fiercely, and then dig in and hold.”



1. Define your Focused Point of Attack

This step is about identifying a “beachhead segment” that you’d use to cross the chasm into the mainstream market. Your goal is to hit the strategic bowling pin that’ll knock down the remaining pins.



In theory, target marketing is simple: you break down the market into possible sub-segments and select the most attractive segment. In reality, people get stuck in decision-paralysis in face of high-stakes decisions with limited information.



Do enough analysis to identify key customer archetypes, e.g. “Gen Y” or “Dinks” (double income, no kids). Then, use scenarios to flesh out your target customer characterization:

- Craft 1-page scenarios to summarize each customer type, including his personal profile and how exactly he’d use your product. Invite people from different departments to contribute. Include the following in each scenario:
 - (i) *Header information* with a description of the (a) end user, (b) technical buyer (who gives advice or installs the product) and (c) economic buyer (who pays for the product). For business clients, include the industry, geography, department and job title. For consumer clients, include the age, gender, economic status and social group.
 - (ii) *Before your solution*: Describe the scene where the user is stuck, including (a) the moment of frustration, (b) the desired outcome, (c) the attempted approach, (d) the obstacles, and (e) the economic consequences.
 - (iii) *After your solution*: Replay the scene with your product, including (a) the new approach, (b) how it helps them to get unstuck and (c) the economic rewards.
- Once you have a library of 1-page profiles, get a small committee to evaluate each option using the Market Development Strategy Checklist. Rate these factors (on a scale of 1-5) in 2 stages:

KEY QUOTES

“Trying to cross the chasm without taking a niche market approach is like trying to light a fire without kindling.”

- (i) *Show-stoppers*. Only consider profiles that score high on all 4 factors.

Target customer	Is there a single, identifiable economic buyer for this offer, whom we can readily access and can afford the price for the whole product?
Compelling reason to buy	Are the economic reasons strong enough for the buyer to want to fix the problem urgently?
Whole products	Can our company (with the help of partners/allies) provide a complete solution in the next 3 mths to satisfy the compelling reason to buy, and dominate the market (>50%) in another 12 mths?
Competition	Has another company already crossed the chasm, addressed this problem and occupies the space?

- (ii) *Nice-to-haves*. Add the scores for 5 more items:

Partners and allies	Do we have existing relationships with other companies needed to fulfill the whole product?
Distribution	Do we have an existing sales channel to reach the target customer and fulfill the whole product requirements?
Pricing	Is the whole product pricing within the target customer's budget and does it justify the value from solving the problem? Can all partners get adequately compensated?
Positioning	Is our company credible for this product and segment?
Next target customer	If we dominate this niche, can it create a "bowling pin effect" to help us enter adjacent niches?

- Shortlist the top options. As a team, **select just 1 segment that you'll commit to. This must be (a) large enough to matter, (b) small enough for you to win, and (c) can fit your strengths.**

- (i) Speed matters. You can commission market research to validate your biggest assumptions, but there's no need for perfect information. You can still succeed even if you didn't pick the optimal choice.
- (ii) To become a leader, you must win >50% of the chosen niche within the 1st year, while you concurrently build new segments. Find a segment that matches your size/goal.
- (iii) If none of your scenarios survive the evaluation, then you're *not* ready to cross the chasm. Stick to early-market projects and keep costs low while you search for a viable beachhead.

KEY QUOTES

"The good news in this is that you do not have to pick the optimal beachhead to be successful. What you must do is win the beachhead you have picked."

"The rules of thumb in crossing the chasm are simple: Big enough to matter, small enough to lead, good fit with your crown jewels."

- In the 1980s, Macintosh successfully crossed the chasm by targeting graphic art departments of Fortune 500 companies. By dominating that segment, they could use it as a beachhead to win over other executives in those companies (for whom the graphic artists provided presentations for), as well as the creative agencies and professionals who worked with the graphic artists.



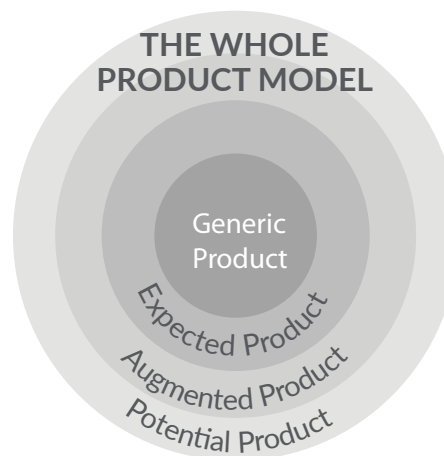
2. Set Up your Strike Force

Define the specific product/offer you'll need to secure your initial target market, and how you'll deliver it with available resources.



Now that you've identified a target market with a compelling reason to buy your product, you must position yourself as the irrefutable choice. To do that, you must deliver a Whole Product (a concept by Theodore Levitt) which comprises 4 parts:

- Generic product = what's in the purchasing contract (e.g. a tablet with cables).
- Expected product = what the consumer thinks he's buying, including the configuration needed to achieve the buying goal (e.g. internet connection needed to use the tablet).
- Augmented product = product extensions that'll maximum the chance of fulfilling the buying goal (e.g. basic software like email, browser, calendar, app store etc.).
- Potential product = future product improvements (e.g. ancillary products or system enhancements).

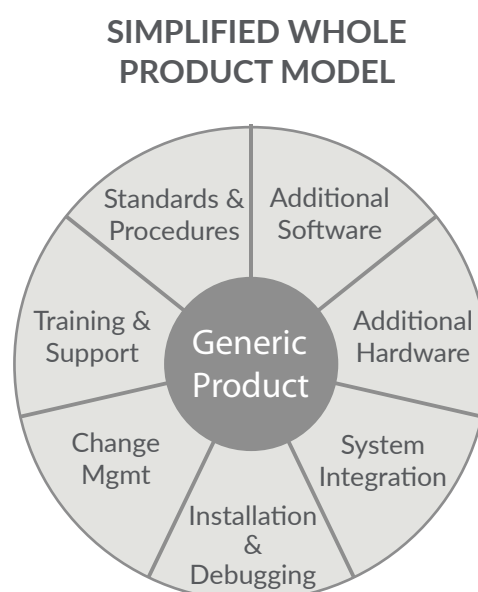


Customers in the early market focus mainly on the generic product, but the focus shifts outward (toward support and ancillary services) in the mainstream market. To achieve market leadership, you must deliver the whole product. Start

KEY QUOTES

with the minimum level of the whole product you'll need to cross the chasm, i.e. what the target customers need to satisfy their compelling reason to buy.

- Use the model on the right to identify generic product vs other components needed to satisfy the buying objective.
- Define which ones will be provided by your company (vs the customers, partners or allies), and ensure all vendors have a fair share of the pie. Invest time/resources to build those relationships.
- Example: Imagine you're replacing an incumbent in providing enterprise wifi for a university. To overcome the inertia to change, your solution must be at least 10x better than the incumbents'. Besides offering the core/generic product (wifi access points throughout the campus and systems for central/network management), you may also provide additional elements such as: (i) building interfaces with old equipment so they can still be used, (ii) partnering with existing campus vendors for a seamless transition, and (iii) customization for other campus initiatives. The additional elements are not strictly required under the contract and will require a lot of extra effort/resources. Yet, they'll help to satisfy the client to win their loyalty and build positive referrals.



KEY QUOTES

"Winning the whole product battle means winning the war."

"The fundamental rule of engagement is that any force can defeat any other force— if it can define the battle."



3. Define the Battle

To dominate the mainstream market, you must know the landscape, who your competitors are, their relationships with your target customers and how to position yourself to win the battle.



Know your target customers' top values and concerns. There're 4 domains involved in marketing any high-tech product: technology product, market and company.

The Competitive Positioning Compass maps out how the 4 factors shift in relative importance as you move down the tech adoption curve:

- Interest & technical know-how. Early markets are driven by specialists who care more about technology and product issues. Mainstream markets are driven by generalists who care more about market leadership and company stability. To cross the chasm, you must shift your focus accordingly.



- Attitude. When people are skeptical, they won't be interested in your product/company. Even if you've already won the support of the visionaries, you must still win over the skeptical pragmatists from scratch.



To win the pragmatists' support, you must effectively position your company/product in their minds. **Positioning** is about firmly associating certain attributes with your product/company, so people will think of you as the incontestable choice in a specific situation. Your unique value proposition must match your target customers, their compelling reason to buy and your product offering.

- It's much easier to fit into people's existing way of thinking than to change them. Pragmatists evaluate options by comparing similar vendors/products in a category. So, you must position your product in a category that the pragmatists are familiar with and have other reasonable choices to choose from. Then, position yourself such that you're the obvious, inevitable choice.
- If there's no direct competition, you must create competition. Help the customers to understand your value proposition with 2 sets of competitive comparisons:

KEY QUOTES

"Crossing the chasm requires moving from an environment of support among the visionaries back into one of skepticism among the pragmatists."

"The goal of positioning is to create and occupy a space inside the target customers' head."

"In general, the most effective positioning strategies are the ones that demand the least amount of change."

"The pragmatists are loath to buy until they can compare. Competition, therefore, becomes a fundamental condition for purchase."

- (i) *Market alternatives* or existing vendors that the target customer has been buying from. Show that you can solve a major limitation in their offer.
- (ii) *Product alternatives* or other tech leaders using a similar disruptive innovation as yours. Refer to their technology to gain credibility but differentiate yourself with a segment-specific focus.

For example, Box explains their value proposition as: “Dropbox’s ease of use meets SharePoint’s enterprise standards.”

- Build credibility in pragmatists’ eyes by presenting evidence of strong partners/allies (e.g. joint-marketing or cross-referencing), market share, certifications, press coverage etc.
- You can only own a tiny space in people’s minds, so it’s crucial to keep your message focused and memorable, to achieve effective word of mouth. Communicate your position in 4 steps:
 - (i) State your *claim* as the indisputable market leader for a specific segment. Condense it into 2 sentences: “For [beachhead target customers] who are [dissatisfied with market alternative], our product is a [product category] that provides [compelling reason to buy]. Unlike [product alternative], we have assembled [whole product features].”
 - (ii) Present the *evidence* to support your claim;
 - (iii) *Communicate* the right message to the right audience; and
 - (iv) Dynamically *refine* your message based on how competitors respond.



4. Launch your Invasion

Your #1 priority when crossing the chasm is to secure a distribution channel to your target market. To achieve that, you need the right pricing and distribution strategies.

KEY QUOTES

“Evaluations are often simply rationalizations of preestablished positioning.”

“As we launch our invasion across the chasm, distribution is the vehicle that will carry us on our mission, and pricing is its fuel.”



Customer-Oriented Distribution. There're 5 main customers types, each with their own priorities and preferred channels. Choose 1 main channel that matches your target segment:

CUSTOMER TYPE	PREFERRED CHANNEL
Enterprise executives (huge purchases for organization-wide systems)	Direct sales: use a consultative approach and tailor the solution to the client's needs. Use relationship marketing to build personal connections with key leaders, before shifting to solution-selling once instructions are given to explore your product.
End users (relatively low-cost purchases for personal or local workgroups)	Web-based self-service: let users find information on the internet and sign up for a free trial before making a purchase. You can also offer a freemium model (where the core technology is provided for free with value-added upsells), or use e-commerce transactions.
Department heads (medium purchases for organizations that need IT systems but lack the budget/inhouse support)	Sales 2.0: Provide direct sales and support over digital media. Provide info on the internet, but have a human salesperson get in touch (via voice call, chat or email) when the client clicks on a link for support. Depending on the interest level, the client may then get directed to website info, a webinar or live demo.
Engineers (decide on technical options on behalf of internal clients)	Two-Tier Distribution: Use the internet to provide data, then have a human contact take them through the options until they're ready to connect you with the purchasing departments.
Small business owners/operators (small purchases that deliver high-value)	Value-added resellers: Use agents (usually sole proprietors) who'll act as low-cost advisers to help the SME owners navigate complex tech requirements and find cheap solutions.

KEY QUOTES



Distribution-Oriented Pricing. Your prices must be low enough to sell yet high enough to be worth selling. Initially, set your pricing at the level of the market leader, so you can (i) reinforce your positioning and (ii) have a large-enough margin to reward your channel partners. Prices will gradually decrease as your product becomes established in the mainstream market and new competition emerges.



Crossing the Chasm Permanently

In the rush to cross the chasm, companies often make unsustainable commitments. Make the right decisions from the onset so you won't fall back into the chasm.



Financial decisions. It is a misconception that high-tech revenues start out flat then curve up sharply like a hockey

stick. In reality, revenues grow like a staircase, with spurts of growth that flatten out each time you enter a new segment.

- Investors should be prepared to sustain the investments for as long as needed to cross the chasm and create a sustainable whole product. Don't just ask for broad business plans—ask to see the detailed battle plans for crossing the chasm.
- So long as the entrepreneurial team depends on investors for cash, they can't control their destiny. The faster you gain financial discipline, achieve profitability and break-even cash flow, the better. When people's salaries come from the market (not investors), they'll also focus more on market needs.



Organizational decisions. To win in early markets, you need pioneers to drive innovations and push new boundaries. However, once you've crossed the chasm, you must reassign the pioneers to new projects and bring in settlers to stabilize and institutionalize your processes/standards for mainstream markets.

- Introduce 2 temporary roles to help with the transition.
 - (i) Once you've secured a visionary account, assign a *target market segment manager* as the account manager. His role is to (a) learn how the business works, (b) manage the project and (c) work with whole product manager to identify essential whole-product components. Once he has completed the 1st phase of the visionary's project, hand the account over to a settler so he can focus on transforming the insights from the visionary client into a potential beachhead segment.
 - (ii) During the chasm period, you must prepare for marketing (even if you don't start marketing yet). Hire a *whole product manager* to take stock of the growing list of bugs and unresolved issues. Once you cross the chasm, make sure R&D focuses on addressing these issues, to deliver a whole product aligned with your mainstream customers, pricing, and distribution channels.

KEY QUOTES

"Until break-even cash flow is achieved, nothing is secure, and your destiny is not under your own control."

"To leave the chasm behind, to cross it and not fall back into it, involves...the move from being pioneers to becoming settlers."

"Whole product R&D is driven not by the laboratory but by the marketplace."

- Compensation. To succeed in mainstream markets, you need much more than an initial product or a few visionary accounts. Don't overcompensate pioneers for early market successes; retain resources so you can reward people who help you to cross the chasm (and stay with you).



Other Details in the Book to Look out For

This is a guide-book for marketing high-tech products, packed with detailed concepts, frameworks and case-studies. Besides the insights and examples in this summary, the book also includes:

- Examples of high-tech products in various industries, e.g. Documentum, Salesforce.com, VMware, Aruba and Lithium;
- Additional principles/insights on positioning, publicity and partnerships; and
- Two appendices that outline (i) the High-Tech Market Development model from the book "Inside the Tornado" and (ii) the Four Gears Model for Digital Consumer Adoption to scale up online adoption.

KEY QUOTES